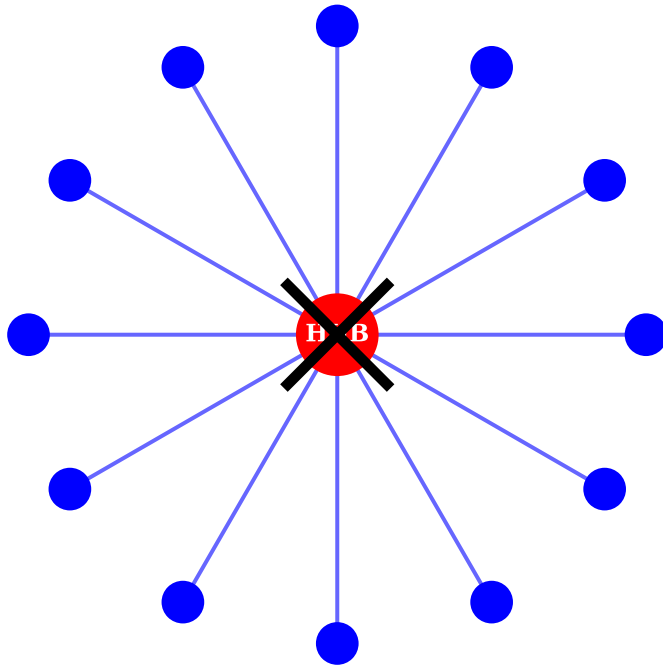


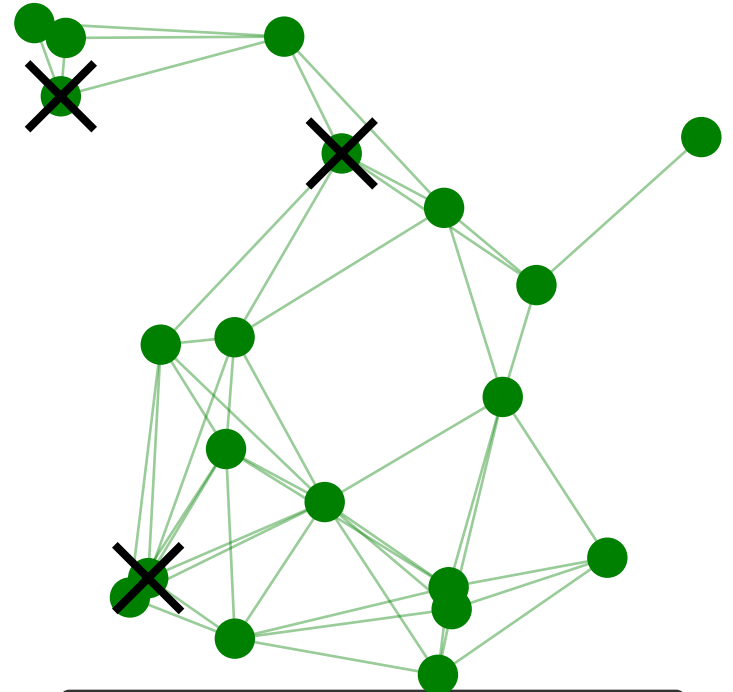
Figure S1: Network Topology Determines Collapse Velocity
The Cascade Amplification Coefficient (λ)

Hub-and-Spoke Network
(High $\lambda \approx 2.0$)



Soviet Union, Bronze Age Palaces
Single point of failure → Catastrophic collapse
 $v_c > 0.05/\text{year}$

Distributed Mesh Network
(Low $\lambda \approx 0.8$)



Roman Empire, Feudal Europe
Redundant pathways → Gradual degradation
 $v_c < 0.005/\text{year}$